Worksheet 1: Lipid Digestion — Concepts and Applications

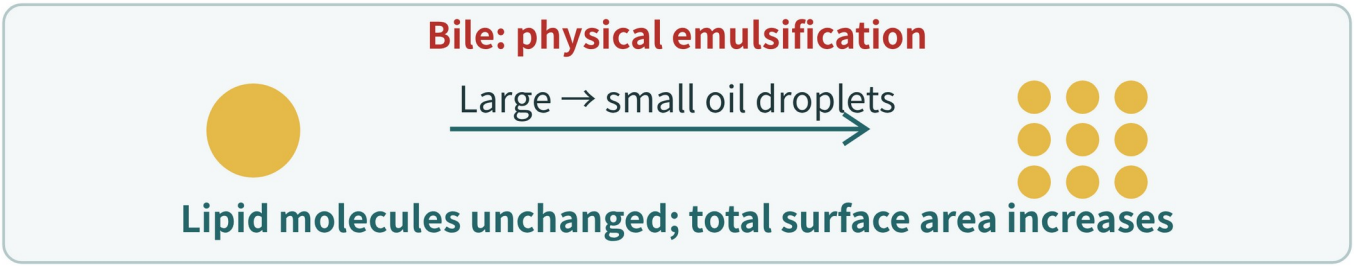
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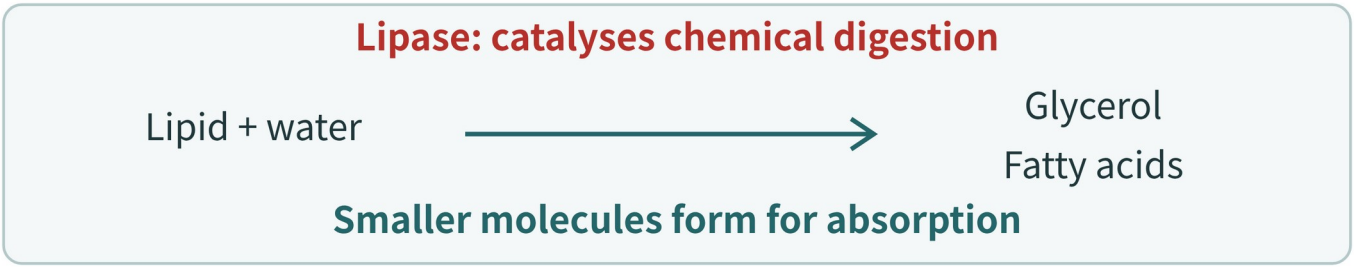
Read the topic notes first, then apply the concepts to analyse the results.

Topic notes

X is **bile**, made in the liver and stored in the gall bladder, containing no digestive enzymes. It **physically emulsifies** large oil droplets into smaller droplets without changing lipid molecules. The greater **total surface area** helps **lipase** contact lipids.

Y is **pancreatic juice** containing **lipase**. **Lipase** catalyses **chemical digestion**, forming **glycerol** and **fatty acids** in this course's simplified complete-hydrolysis model. **Lipase** is a **protein**: boiling **denatures** it and stops its action, which cooling does not restore. **Bile** retains emulsifying action after boiling in this model.





Context: lipids need digestion before absorption. What happens after adding digestive juice X or Y?



Part 1 | Main investigation design: compare X and Y

Add 2 mL X or 2 mL Y to identical oil–water mixtures. Shake in the same way at 37°C; compare after one hour.

1. Identify the variables: choose all appropriate factors in each row.

IV(the factor changed on purpose)；DV(the result you measure or observe)；CV(the factors kept the same)

Independent variable	Total volume of oil and water ／ Reaction temperature ／ Reaction time ／ Total volume of liquid added ／ Method of shaking ／ Appearance of the mixture ／ Combination of digestive juices
Dependent variable	Total volume of oil and water ／ Reaction temperature ／ Reaction time ／ Total volume of liquid added ／ Method of shaking ／ Appearance of the mixture ／ Combination of digestive juices
Controlled variables	Total volume of oil and water ／ Reaction temperature ／ Reaction time ／ Total volume of liquid added ／ Method of shaking ／ Appearance of the mixture ／ Combination of digestive juices

2. Select the assumptions for a fair comparison (choose all that apply)

(An assumption is a condition we have not proved but take as true, to help make the experiment results valid.)

□ Observe all setups after the same reaction time.

□ Use the same starting amounts of oil and water and the same total volume of added liquid in each setup.

□ Use the same method of shaking for every setup.

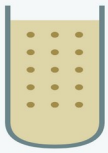
□ The X setup may receive more total liquid than the Y setup.

Part 1 | Apply concepts to the individual actions of X and Y

Both model results show one hour after adding the juice and shaking.

Set X

Add 2 mL X



Cloudy emulsion

Set Y

Add 2 mL Y



Thinner oil layer; two layers remain

3. Appearance of X after one hour

☐ a cloudy emulsion

☐ The oil layer disappears

4. Appearance of Y after one hour

☐ Thinner oil layer, with two layers remaining

☐ No obvious change in the oil layer

Apply emulsification and surface-area concepts to analyse how X helps Y.

Extension 1 uses two fresh tubes. Add X or an equal volume of water and shake, then add the same amount of Y. Both have the same pretreatment time.

Part 2 | Extension 1: how does X help Y?

The following model results show one hour after adding Y.

X first, then Y

1 mL X → 1 mL Y



Less cloudy
No obvious oil layer

Water first, then Y

1 mL water → 1 mL Y



Thinner oil layer
Two layers remain

5. How does X help Y act?

X disperses large oil droplets into smaller droplets. The total surface area of the same amount of lipid:

☐ increase

☐ decrease

Part 2 | Extension 1: further analysis

6. What does appearance tell us?

□ Appearance provides clues about the effects of digestive juices, but alone cannot prove that glycerol and fatty acids have formed.

□ Disappearance of the oil layer completely proves that lipids have broken down into glycerol and fatty acids.

Part 3 | Extension 2: effects of boiling

Use unboiled X followed by Y from Extension 1 as the baseline. Boil one juice, cool to 37°C, then add 1 mL each of X and Y in the same order; observe after one hour.

Boiled X + unboiled Y

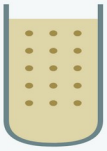
Use after cooling



Less cloudy; no oil layer

Unboiled X + boiled Y

Use after cooling



Cloudy emulsion

7. Which juice loses its action after boiling?

□ Digestive juice X

□ Digestive juice Y

Integrated explanation

8. Using the experimental results and topic notes, explain how bile and lipase in pancreatic juice work together to help digest lipids, and how boiling affects their actions.